Khalil Sahli

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Machine Learning Engineer (GenAI, LLM agents, edge AI)

Education

The National Engineering School of Carthage (Enicarthage) — Charguia 2, Tunis
Computer Engineering Degree – 3-year engineering cycle, Master's-equivalent — *September 2024 – May 2027 (expected)*

Tunis Preparatory Engineering Institute (IPEIT) — Montfleury, Tunis
Preparatory Classes for Engineering Schools – 2 years, Mathematics & Physics — *September 2022 – May 2024*

Experience

Generative AI Engineer (Intern) — June 2026 – August 2026
STMicroelectronics — *El Ghazela, Tunis*
- Designed the chunking and retrieval structure feeding an internal RAG system, then benchmarked it on 50 questions across 25 errata sheets, comparing flat BM25, metadata-scoped BM25 and structured search
- Built eight Python packages that turn technical PDFs into structured JSON, recovering 178 tables and 930 sections
- Guarded the extraction rules with 928 automated tests and a citation check that reopens each cited page

Software Engineer (Intern) — July 2025 – August 2025
Sagemcom — *Ez Zahra, Tunis*
- Cut manual remediation effort by 80% by automating Wi-Fi card corrections with idempotent update logic
- Built a Python tool that parses device-enclosure logs against a canonical reference and emits CSV reports
- Kept releases reliable across heterogeneous hardware by adding unit tests and CI pipelines with pytest

Machine Learning Engineer (Student Researcher) — October 2024 – November 2024
IEEE Computer Tunisia Chapter — *Soukra, Tunis*
- Held object recognition accuracy steady across changing lighting and scene conditions by building a CNN perception pipeline for real-time detection and classification from camera input, backed by a data augmentation stage
- Benchmarked detection architectures and loss functions to maximize mAP and inference speed on GPU hardware
- Cut model iteration time by scripting reproducible training and evaluation workflows in PyTorch and OpenCV

Projects

TicketPilot – AI Agent with RAG & Tool Calling | *Python, LangGraph, Qdrant, FastAPI* — Apr 2026 – May 2026
- Built an AI agent that classifies each request, retrieves the applicable rule, calls business tools through REST APIs and drafts a cited answer: 45% of 150 labeled cases resolved automatically at 97.1% precision, the rest routed to human review
- Cut unsupported claims from 11.3% to 2.7% with a verification node; hard-coded guardrails blocked 12/12 prompt injections
- Lifted recall@5 from 72.0% (BM25) and 78.7% (embeddings) to 86.0% with hybrid vector search (RRF) in Qdrant

Natural Language to Circuit | *Python, C++, CodeLlama-13B, LoRA, ngspice* — Jan 2026 – May 2026
- Turned a plain-language description into a simulated circuit by chaining two LoRA fine-tuned CodeLlama-13B models that write SPICE netlists and schematics, then driving ngspice to verify the result
- Rendered the generated circuits as diagrams through a custom C++ automatic diagram-generation engine
- Exported LTspice-compatible schematics alongside ngspice simulations, openable in standard industry tools

CircuitVision – Detection, Segmentation & VLM | *Python, PyTorch, YOLOv11, SAM 2, Docker* — Jan 2026 – Feb 2026
- Turned photos of circuit diagrams into simulation-ready SPICE netlists at 0.93 mAP-50, by fine-tuning YOLOv11 for component detection and chaining it with SAM 2 segmentation and a vision-language model that reads component labels
- Reached 98.7% segmentation accuracy on 267 annotated images by adapting SAM 2
- Removed local GPU dependency by containerizing the pipeline with Docker and serving it from RunPod

TinyML CAN Bus Intrusion Detection | *Python, scikit-learn, C, X-CUBE-AI, STM32F446* — Sep 2026 – Oct 2026
- Detected DoS and four spoofing attacks on a live 500 kbit/s CAN bus with an int8 model deployed on an STM32F446 via X-CUBE-AI: 40 µs per frame, 20 KB flash, 1 KB RAM, macro-F1 0.94 vs 0.95 in float
- Built a leakage-free evaluation on CICIoV2024 (1.4M real-vehicle frames): removed the duplicates that inflate macro-F1 to 0.99, weighted rare classes, 0.1% false positives and 0.5 false alarms per minute live
- Benchmarked a random forest against a compact MLP and deployed the MLP: 1.1 KB of weights, 25 µs per inference

VHDL Neural Network – FPGA Digit Recognition | *VHDL, PyTorch, ModelSim, Spartan-6* — Sep 2026 – Oct 2026
- Ran a 64-32-10 MLP entirely in hardware: trained it in PyTorch to 97.50% test accuracy, quantized every weight to 8-bit Q2.6 with no loss against the float model, and derived the accumulator widths (15 and 19 bits) that make overflow impossible
- Built the datapath from gates in VHDL (array multipliers, a 6-level adder tree, ReLU shift-saturate, argmax tree) and verified it bit-exact against the Python reference on all 360 test images
- Deployed a MAC-based variant on a Nexys 3 Spartan-6 board, driven over USB-UART from a PC sketchpad

Technical Skills

Generative AI & LLMs: RAG, AI agents, LangGraph, LoRA fine-tuning, Hugging Face, Qdrant, Gemini API, Whisper
Machine Learning & Vision: PyTorch, TensorFlow, scikit-learn, YOLOv11, SAM 2, OpenCV, model evaluation
Edge AI & Embedded: int8 quantization, X-CUBE-AI, STM32, FreeRTOS, C, VHDL, FPGA
Software & Tools: Python, C++, SQL, FastAPI, Docker, Git, CI/CD, pytest, Linux, Windows, GCP
Languages: Arabic (native), English (B2), French (B2), Spanish (basic), Japanese (basic)